# ha-coap-node Rev 2.0 Schematic Review

Review of the nRF52840 soil-moisture and pump node schematic, cross-checked against component datasheets and the `ha-coap-server` firmware

| | |
|---|---|
| **Board** | ha-coap-node, Prototype variant, Rev 2.0 (schematic dated 2024-02-20, author Yann T., Backburner Labs) |
| **Reviewed document** | Schematic_ha-coap-node-Rev-2.0.PDF (one sheet). Connections were read from the PDF's embedded netlist. |
| **Firmware context** | `ha-coap-server` at commit `04789f2` (board target `hacoap`) |
| **Review date** | 2026-10-24 |
| **Scope** | Schematic and netlist only. PCB layout, RF and antenna keep-out, BOM sourcing and mechanical design were not reviewed. |
| **Prepared by** | Claude (AI-assisted review, prepared with Claude Code). Items marked *verify on hardware* should be confirmed on a built board. |

## Summary

The architecture is sound: a proper USB/battery power path, the fuel gauge on the battery side, power-gated sensor supply and well-strapped sensors. However, **the power-off path cannot work as drawn**. The pushbutton, the MCU's PWR_KILL line and the 3.18 V low-battery cutoff all depend on the TPS2115A turning its output off, and R5 holds its D1 input low permanently. Two other functional issues need fixing: the pump has no freewheeling diode, and the charger's BAT pin has too little capacitance.

For a sleepy end device on battery, the quiescent current is dominated by passive parts rather than the radio. These include an always-on power LED (about 1 mA) and a 10 kΩ pull-up that draws 300 µA whenever SW2 is set to MTD, which is the battery mode.

| ID | Severity | Finding | Recommended change |
|---|---|---|---|
| F1 | HIGH | Power-off path is inoperative. R5 pulls TPS2115A D1 to GND, but the LTC2950-2 EN output is open-drain, so D1 can never go high. | Do not fit R5. Also stop deriving D0 from OUT (do not fit R4). See F1. |
| F2 | HIGH | No freewheeling diode across the pump (J2). D1 sits across VCC–GND instead. | Add a Schottky diode across J2 (cathode to VCC). |
| F3 | MEDIUM | The STC4054 BAT pin has only 100 nF, and the battery is on a flying lead. | Add 4.7 µF near U4 BAT. |
| P1 | HIGH | LED1 power LED is always on (about 1 mA). | Do not fit it, or switch it with a jumper or GPIO. |
| P2 | HIGH | R36 draws 300 µA whenever SW2 is ON, and ON = MTD (the battery mode). | Reverse the switch sense, or use a 1 MΩ pull-up. |
| P3 | MEDIUM | LED3 on ALERTn draws about 1.2 mA exactly when the battery is low. | Use a high-value pull-up and drive the LED from a GPIO. |
| P4 | MEDIUM | R30 (XSHUT pull-up) draws 300 µA whenever XSHUT is held low. | Keep XSHUT high, or raise R30 to 100 kΩ or more. |

| ID | Severity | Finding | Recommended change |
|---|---|---|---|
| P5 | MEDIUM | The TPS2115A draws 55 µA typical (90 µA max) while running. | Budget for it, or use a lower-current power multiplexer. |
| P6 | LOW | The KILL divider draws about 10 µA from the battery at all times. | Scale the resistors up 2–3×. |
| M1 | MEDIUM | CHRGn (P1.10) is pulled toward VUSB through LED4 and goes above the MCU's 3.3 V pin limit. | Add a series resistor or a divider to the MCU pin. |
| M2 | MEDIUM | USB_PWR (P0.15) floats on battery because nothing pulls it up to the 3.0 V rail. | Add a pull-up to VMCU (the internal one is fine). |
| M3 | MEDIUM | The Q1 gate threshold can be 2.5 V, and on-resistance is only specified at 4.5 V and 10 V. The MCU drives the gate at 3.0 V. | Use a MOSFET specified at $V_{GS}$ = 2.5 V. |
| M4 | MEDIUM | Battery sag when the pump starts can trip KILL (30 µs minimum pulse) once F1 is fixed. | Add a capacitor across R35. |
| M5 | MEDIUM | No battery temperature check before charging, on an outdoor device. | Inhibit charging in firmware via PROG, or use a charger with a thermistor input. |
| M6 | LOW | 500 mA charge current: heat in a SOT-23, the USB current budget and the battery C-rate. | Confirm against the battery capacity. |
| M7 | LOW | The soil sensor cable on J5 is exposed outdoors and has no TVS protection. | Add a TVS diode and raise R25. |
| L1–L8 | LOW | Documentation errors and small cleanups. | See the minor and documentation section. |

## 1. Functional issues

### F1 The power-off path cannot turn the board off HIGH

**Circuit.** U9 (LTC2950CTS8-2) pin 6, EN-bar, drives net SW_D1, which goes to U2 (TPS2115A) pin 3, D1. R5 (10 kΩ) pulls SW_D1 to GND. U2 pin 2 (D0, net SW_D0) is pulled up to +VCC by R4 (10 kΩ). +VCC is U2's own output, OUT.

**Datasheet behavior.** The LTC2950-2 EN-bar is an **open-drain** output. It is driven low when on and released (high-impedance, at most 0.1 µA leakage) when off. On the TPS2115A, the only way to switch OUT off is D1 = 1 with D0 = 1. D0 and D1 each have an internal pull-up of about 1 µA.

| D1 | D0 | $V_{IN2} > V_{IN1}$ | OUT | STAT |
|---|---|---|---|---|
| 0 | 0 | X | IN2 | Hi-Z |
| 0 | 1 | No | IN1 | 0 |
| 0 | 1 | Yes | IN2 | Hi-Z |
| 1 | 0 | X | IN1 | 0 |
| 1 | 1 | X | Hi-Z | 0 |

**Result.** When EN-bar is released, R5 (10 kΩ) easily outweighs D1's 1 µA pull-up, so D1 stays at 0 in both states. The output never turns off. The pushbutton (SW1), the MCU's PWR_KILL (P0.04 through Q4) and the low-battery cutoff (R31/R35 on KILL) therefore all switch the LTC2950 without any effect. The battery can discharge until its own protection circuit cuts it off. At power-up, D0 starts low because OUT is 0 V, so IN2 (battery) is selected. When OUT rises, D0 goes high and the part enters automatic mode. So normal operation works, which may have hidden this problem. *Verify on hardware: press and hold SW1 and check whether the board ever powers off.*

**Why fixing R5 alone is not enough.** If D1 can go high but D0 still follows OUT through R4, an "off" board with USB connected would cycle repeatedly. D1 = 1 and D0 = 0 selects IN1, so OUT rises. D0 then goes high, which switches OUT off. OUT decays, D0 falls, IN1 is selected again, and the MCU would brown out on every cycle.

**Recommended change.**

- **R5:** do not fit. D1's internal pull-up holds it high when EN-bar is released, and EN-bar's leakage is only 0.1 µA. Alternatively, fit a pull-up of 1 MΩ or more instead of a pull-down.
- **R4:** do not fit (R34 stays unfitted), so D0 is held high by its internal pull-up. Do not take D0 from OUT.
- Resulting behavior: **on** (EN-bar low) gives D1 = 0, D0 = 1, automatic selection of the higher of USB and battery. **Off** gives D1 = 1, D0 = 1, output off. The battery still charges while the board is off, because U4 runs directly from VUSB.
- Design decision: U9 is powered only from VBATT. With no battery fitted, the board stays off even with USB connected. If USB-only operation is required, rethink both U9's VIN (for example a diode-OR of VUSB and VBATT) and the KILL divider, which would hold KILL low when no battery is fitted.

### F2 No freewheeling diode across the pump HIGH

**Circuit.** J2-1 is +VCC and J2-2 is the Q1 (TSM850N06CX) drain, which is a net containing only J2-2 and Q1 pin 3. D1 (SD05T1G) and C6 are both connected between +VCC and GND, not across the load.

**Impact.** When Q1 turns off, the motor's inductive kick at the drain has no current path. The voltage rises until Q1 (rated 60 V) breaks down, on every pump cycle. This stresses Q1 and couples noise onto VCC, which also feeds the LDO for the MCU rail.

**Recommended change.** Add a Schottky diode directly across J2, with the cathode on J2-1 (VCC) and the anode on J2-2 (drain), rated for the pump current or more (for example 1 A in SOD-123F). D1 can stay as a supply clamp.

### F3 Too little capacitance on the charger's battery output MEDIUM

**Circuit.** The only capacitors on +VBATT are C8 (100 nF, at U2 IN2) and C15 (100 nF, at U9 VIN). The battery connects through J1 (JST-SH) on a wire lead.

**Datasheet.** STC4054 section 6.13: lead inductance may require at least 1 µF from BAT to GND. A 4.7 µF capacitor with 0.2–1 Ω in series is required to keep ripple low when the battery is disconnected. ST's own characterization used about 4.7 µF.

**Recommended change.** Add 4.7 µF (X5R/X7R, 10 V or higher) close to U4 pin 3, with the recommended series resistance or a capacitor with matching ESR.

## 2. Power consumption

These loads are present in normal battery operation. Each is large compared with what the nRF52840 radio averages as a Thread sleepy end device with a 1 s poll period.

| ID | Load | When it draws current | Current | Recommendation |
|---|---|---|---|---|
| P1 | LED1 green power LED, through R2 1 kΩ from VMCU | Always | ≈1 mA | Do not fit for battery builds, or switch it with a jumper or GPIO. A high-value resistor (47–100 kΩ) gives a faint indicator. |
| P2 | R36 10 kΩ pull-up on GPIO_0_24 (SW2) | While SW2 is ON, and the drawing defines ON = MTD | 300 µA | Reverse the sense so that ON = FTD (normally USB-powered), or use a 1 MΩ pull-up, or use the MCU's internal pull-up only while reading the switch at boot. |
| P3 | LED3 red, through R8 1 kΩ on ALERTn (MAX17048 ALRT is open-drain) | While a fuel-gauge alert is active, which is typically low battery | ≈1.2 mA | Use a 100 kΩ pull-up on ALERTn and drive any LED from a GPIO. The LED currently drains the battery fastest exactly when the battery is lowest. |
| P4 | R30 10 kΩ pull-up on TOF_SHNn (VL53L0X XSHUT) | While the MCU holds XSHUT low (current firmware does) | 300 µA | Hardware standby and software standby on the VL53L0X draw nearly the same current, so keep XSHUT high. If MCU control is needed, raise R30 to 100 kΩ–1 MΩ. |
| P5 | U2 TPS2115A operating supply current | Always while powered | 55 µA typ<br>90 µA max | Budget for it. If battery life is a priority, evaluate a power multiplexer with single-digit-µA quiescent current. |
| P6 | R31 330 kΩ + R35 76.8 kΩ KILL divider from VBATT | Always, even when the board is off | ≈9–10 µA | Scale up about 2–3× (for example 1 MΩ / 232 kΩ). KILL leakage is at most 0.1 µA, so the threshold error stays under about 3%. |
| — | U9 LTC2950 (6 µA typ), U1 MAX17048 (about 3 µA in hibernate, about 25 µA active, per the drawing's notes), sensors in standby | Always | ≈10 µA | Acceptable. Listed for completeness. |

> **Related, outside the schematic:** the ACN52840 module includes the nRF52840's DC/DC converter components, but the `hacoap` board definition does not enable the DC/DC regulator. Enabling it roughly halves radio current.

## 3. Medium-priority items

### M1 CHRGn (P1.10) can exceed the MCU's input limit MEDIUM

STC4054 CHRG has three states: a strong pull-down while charging, a weak pull-down of about 20 µA when charging is done or the charger is in standby, and high-impedance when there is no input power. On this board CHRG is pulled toward VUSB (5 V) through LED4 and R7. In the "done" state the node settles about one LED drop below VUSB (≈3.3–3.5 V). That exceeds the nRF52840's VDD + 0.3 V limit on a 3.0 V rail, and it back-feeds VMCU through the pin's ESD diode when the board is off with USB connected. LED4 will also glow faintly when charging is done.

**Change:** feed P1.10 through a series resistor (47 kΩ or more) or a divider, or move the LED to its own path and sense CHRG with the MCU's internal pull-up.

### M2 USB_PWR (P0.15) has no pull-up on the battery side MEDIUM

TPS2115A STAT is open-drain. It is low when IN1 (USB) is on or the output is off, and high-impedance when IN2 (battery) is on. Its only pull-up is LED5 and R6 to VUSB, which is 0 V when running on battery, so P0.15 floats exactly when it should read "on battery". The yellow LED therefore means "running on USB".

**Change:** add a pull-up to VMCU. The MCU's internal pull-up is sufficient, since it only draws current while on USB power.

### M3 Pump MOSFET gate drive margin MEDIUM

The Q1 TSM850N06CX gate threshold is 1.2–2.5 V (1.8 V typical). On-resistance is only specified at $V_{GS}$ = 4.5 V (100 mΩ max) and 10 V. The gate is driven from P0.17 at 3.0 V, so a worst-case part is barely enhanced, and dissipation rises with pump inrush or stall current.

**Change:** choose an N-MOSFET whose $R_{DS(on)}$ is specified at $V_{GS}$ = 2.5 V. A 20–30 V rating is enough for this rail.

### M4 Low-battery cutoff: sensitivity to pump surges and threshold MEDIUM

The KILL threshold is 0.6 V falling with 30 mV typical hysteresis. Through R31/R35 that gives about **3.18 V falling and 3.34 V rising** at the battery. KILL responds to pulses as short as 30 µs. Once F1 is fixed, battery sag at pump start (battery resistance, the JST-SH contacts and the TPS2115A's 84 mΩ) could trip a shutdown well above the real cutoff voltage.

**Change:** add 100 nF–1 µF across R35. With the present values that is a time constant of about 6–60 ms. The 512 ms turn-on blanking already covers start-up.

### M5 No battery temperature check before charging MEDIUM

The STC4054 has no thermistor input. Charging a lithium-polymer cell below about 0 °C damages it, which is a realistic condition for an outdoor garden device. The datasheet documents PROG as a shutdown pin: pulling it above 1.21 V (typical) disables the charger. P0.05 (I_BATT) can therefore inhibit charging when the HDC1080 reads cold, as long as firmware treats this as intentional (see firmware cross-references).

### M6 Charge current, heat and USB budget LOW

- R13 = 2 kΩ sets 500 mA. Dissipation in U4 is up to (5.0 V − 3.5 V) × 0.5 A ≈ 0.75 W in a SOT-23-5, so the 120 °C die limit will reduce the current. Keep the HDC1080 thermally away from U4.
- Confirm that 500 mA is at most 1C (preferably 0.5C) for the ASR00008 cell.
- Charging (500 mA) plus the pump (200 mA) plus the system can exceed a standard 500 mA USB 2.0 port.

### M7 Soil-sensor connector J5 is exposed outdoors LOW

The J5 cable runs into soil outdoors, but there is no ESD or surge protection on J5-2 (+VSENSOR) or J5-3 (signal). VSENSOR comes from VCC when SEL is low, which is up to 5.25 V on USB. If a sensor ever outputs more than 3.3 V, R25 (1 kΩ) allows up to about 2 mA into AIN0's protection diode.

**Change:** add a low-capacitance TVS on J5-2 and J5-3, and consider raising R25 to 10 kΩ. With C20 = 100 nF the time constant is 1 ms, well inside firmware's 200 ms settling delay.

## 4. Minor and documentation items

| ID | Item |
|---|---|
| L1 | **FPF1320 note is backwards.** The drawing says "SEL = 1 → VOUT = VINA". Per the FPF1320 datasheet, SEL low selects VINA and SEL high selects VINB. U10 VINA is +VCC and VINB is +VMCU, so VS_SEL = 0 selects VCC. The firmware already assumes this. |
| L2 | **Low-battery shutoff formula** references R32. It should be R31: V = 0.6 V × (R31 + R35) / R35 ≈ 3.18 V typical (the drawing states 3.205 V). |
| L3 | The block title "PWR SUPERVISER" should be "PWR SUPERVISOR". |
| L4 | LED2 (blue) is driven from a 3.0 V GPIO through 1 kΩ. Blue LED forward voltage is often 2.7–3.1 V, so brightness will be low and vary between parts. Check the LED's forward voltage at 1 mA, or reduce R3. |
| L5 | J6 (SWD) has no VTref or target-voltage pin, which J-Link and Nordic DK "debug out" need. It is currently unfitted (R26–R28 not fitted), and the ACN52840 has its own Tag-Connect SWD footprint. Either remove J6 or add VMCU to it. |
| L6 | J1 (battery) has no reverse-polarity protection. JST-SH battery polarity is not standardized between vendors. |
| L7 | Mounting holes M1–M4 are connected to each other but not to GND. Confirm this is intentional. |
| L8 | D1 (SD05T1G, 5.0 V working voltage) sits on a rail that reaches about 5.25 V on USB. Check its leakage at that voltage. |

## 5. What is solid

- Proper power path: the charger feeds the battery directly while the TPS2115A powers the system from the higher source. Charging is independent of system load.
- Q2 disconnects $R_{PROG}$ when there is no USB, and PROG is routed to an ADC input (I_BATT), giving a measurement of charge current.
- The MAX17048 is powered from the battery side, so it keeps tracking while the board is off. CTG and QSTRT are grounded per the datasheet, and VDD and CELL are decoupled.
- The VL53L0X has 4.7 µF + 100 nF decoupling per ST. The LSM6DSM is strapped for $I^2C$ at 0x6A (CS high, SA0 low, SDx/SCx grounded). The HDC1080's DAP and NC pins are handled correctly.
- No $I^2C$ address conflicts: 0x36, 0x40, 0x29 and 0x6A. The 4.7 kΩ pull-ups suit a 3.0 V bus.
- USB has an ESD array on VBUS, D+ and D−, the ID pin is left unconnected, and R15 bleeds VUSB down.
- Every MOSFET gate has a 47 kΩ pull-down, so outputs are defined during reset. The buttons have RC debounce.
- Sensor supply is gated by the FPF1320. "Low frequency I/O" notes on the pins near the radio show RF awareness.
- The KILL divider plus Q4 combines a low-battery lockout with an MCU-initiated power-off on one pin (once F1 is fixed).

## 6. Firmware cross-references

These are not schematic defects. They are places where the current firmware (`ha-coap-server` at `04789f2`) interacts with the hardware above.

| Location | Observation |
|---|---|
| `boards/arm/hacoap/hacoap-pinctrl.dtsi` | UART0 RTS is assigned to **P0.05 (I_BATT / STC4054 PROG)**. Zephyr drives RTS high when the UART starts, which holds PROG at about 3.0 V, above the 1.21 V shutdown threshold, so **the charger is disabled**. CTS is assigned to P0.07, which is not on the module. Remove both. *Verify on hardware that charging currently works.* |

| Location | Observation |
|---|---|
| `hacoap board Kconfig` | The DC/DC regulator is not enabled, even though the module has the DC/DC components (see section 2). |
| `dk_leds_init() / tof_en` | P0.09 (XSHUT) is driven low at boot, into R30, and also has `GPIO_PULL_UP` set in the devicetree. See P4. |
| `hacoap.dts` | The VL53L0X is declared at 0x30, but the hardware default address is 0x29 (0x52/0x53 on the drawing), and there is no `xshut-gpios` property. This only matters once time-of-flight code is enabled. |
| `hacoap.dts` | P0.15 USB_PWR is not described (it needs a pull-up, see M2). P0.04 PWR_KILL, P0.31 PWR_BTN_INTn, P0.30 V_BAT, P0.10 TOF_INT and P0.03 AN1 are unused. P0.24 (SW2), P1.09 ALERTn and P1.10 CHRGn are defined but never read. |
| `src/coap_server.c` | The VS_SEL polarity (0 = VCC) matches the FPF1320 datasheet. The schematic note is what's wrong (L1). |

## Appendix A: MCU pin map

U5 ACN52840 (nRF52840 module). Module pins 12, 24 and 32 are VDD (+VMCU, 3.0 V). Pins 1, 13 and 35 are GND. The module also carries an RGB LED, both crystals, a DC/DC converter and a Tag-Connect SWD footprint.

| Pin | nRF52840 | Net | Connects to | Firmware use |
|---|---|---|---|---|
| 5 | P0.02 / AIN0 | AN0 | R25 1 kΩ / C20 100 nF, then J5-3 (soil sensor) | ADC channel 0 |
| 6 | P0.03 / AIN1 | AN1 | J4-9 | unused |
| 14 | P0.04 / AIN2 | PWR_KILL | Q4 gate (R23 47 kΩ), which pulls U9 KILL low | unused |
| 11 | P0.05 / AIN3 | I_BATT | U4 PROG, and R13 2 kΩ to GND through Q2 (on when VUSB is present) | UART0 RTS (conflict) |
| 17 | P0.06 | UART_TX | J4-6 | UART0 TX |
| 20 | P0.08 | UART_RX | J4-7 | UART0 RX |
| 19 | P0.09 / NFC1 | TOF_SHNn | U7 XSHUT, R30 10 kΩ pull-up | tof_en (driven low) |
| 18 | P0.10 / NFC2 | TOF_INT | U7 GPIO1, R29 10 kΩ pull-up | unused |
| 22 | P0.12 | BUZZER | Q3 gate (R21 47 kΩ), drives BZ1 from VCC | PWM0 channel 0 |
| 27 | P0.13 | LED_BLUE | R3 1 kΩ to LED2 | user LED (pump active) |
| 28 | P0.15 | USB_PWR | U2 STAT; LED5/R6 to VUSB | unused |
| 29 | P0.17 | PWR_SW | Q1 gate (R1 47 kΩ), drives the pump on J2 | pump1 |
| 30 | P0.24 | GPIO_0_24 | SW2 (FTD/MTD select, ON = GND), R36 10 kΩ pull-up, J4-8 | defined, unused |
| 15 | P0.26 | SDA | U1, U6, U7, U8; R10 4.7 kΩ; J4-4; TP3 | $I^2C0$ SDA |
| 16 | P0.27 | SCL | U1, U6, U7, U8; R9 4.7 kΩ; J4-5; TP2 | $I^2C0$ SCL |
| 7 | P0.28 / AIN4 | SENSOR_EN | U10 FPF1320 EN | sensor-en |
| 8 | P0.29 / AIN5 | IMU_INT1 | U8 LSM6DSM INT1 | IMU interrupt (driver) |
| 10 | P0.30 / AIN6 | V_BAT | R32 (not fitted) / R33 divider from VBATT | unused |
| 9 | P0.31 / AIN7 | PWR_BTN_INTn | U9 INT, R22 10 kΩ pull-up | unused |
| 21 | P1.09 | ALERTn | U1 ALRT; LED3/R8 to VMCU | defined, unused |
| 2 | P1.10 | CHRGn | U4 CHRG; LED4/R7 to VUSB | defined, unused |
| 3 | P1.13 | BTN | S1 to GND, R11 10 kΩ pull-up, C3 | user button (runs pump) |
| 4 | P1.15 | VS_SEL | U10 FPF1320 SEL (0 = VCC, 1 = VMCU) | sensor-pwr |
| 31 | P0.18 / RESET | RESET | S2, R12 10 kΩ, C4; R26 (not fitted) to J6 | pin reset |
| 33 / 34 | SWDIO / SWDCLK | SWDIO / SWDCLK | R27 / R28 (not fitted) to J6 | — |
| 23 / 25 / 26 | VBUS / D− / D+ | +VUSB / D_N / D_P | J3 micro-USB through the D2 ESD array | USB device (CDC ACM) |

### Power tree and I²C map

- **+VUSB** (J3) feeds U4 STC4054 (500 mA charge), which charges **+VBATT** (J1, battery ASR00008).
- +VUSB (IN1) and +VBATT (IN2) feed U2 TPS2115A (current limit 500/402 Ω ≈ 1.24 A), which produces **+VCC**. +VCC feeds U3 S-1318 (3.0 V LDO), which produces **+VMCU**.
- +VCC feeds the pump (J2, Q1), the buzzer (BZ1, Q3), J4-1, and FPF1320 VINA. +VMCU feeds FPF1320 VINB. The FPF1320 produces **+VSENSOR** (J5-2).
- The following are always on from +VBATT: U1 MAX17048 (through R18), U9 LTC2950 VIN, and the R31/R35 KILL divider.
- I²C addresses (7-bit): MAX17048 0x36, HDC1080 0x40, VL53L0X 0x29, LSM6DSM 0x6A.

## References


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7. Same Sky (CUI) CPT-7502-65-SMT-TR: rated 3 Vp-p, 6 kHz, 1–25 Vp-p operating range. https://sameskydevices.com/product/audio/buzzers/audio-transducers/cpt-7502-65-smt-tr